



**UGANDA INSTITUTE OF INFORMATION AND COMMUNICATIONS
TECHNOLOGY**

FINAL YEAR PROJECT REPORT

KACITA ONLINE TRANSPORT BOOKING SYSTEM

BY

STUDENT NAME	REGISTRATION NUMBER

PROJECT CODE:

DEPARTMENT OF ICT AND ENGINEERING

SUPERVISOR

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UGANDA INSTITUTE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

A PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF INFORMATION AND COMMUNICATIONS TECHNOLOGY AND ENGINEERING IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF A DIPLOMA IN OF UGANDA INSTITUTE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY.

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Final Project Report

Refers to a report compiled that adds to the existing body of knowledge in the respective discipline. The tense used for writing in this document is past tense. At UICT, the project report consists of five sections as follows:

- i. Preliminary pages;
- ii. Chapter 1: Introduction
- iii. Chapter 2: Literature review
- iv. Chapter 3: Methodology
- v. Chapter 4: Systems Analysis, Design and Development
- vi. Chapter 5: System Testing and Validation
- vii. Chapter 6: Limitations, Challenges and Recommendation
- viii. References
- ix. Appendices (Questionnaires, interview guides, program code..e.tc)

The final project report should not exceed 30 pages (and not less than 25 pages) **excluding** title page, table of contents, list of tables and figures, list of abbreviations and appendices but including the list of references appendices. The proposal is submitted to the Projects Office for evaluation and final approval and recording.

The Report should be prepared in Times New Roman Font size 12, 1.5 line spacing.

Preliminary pages

Front Hard Cover of a Final Year Project Report

The front hardcover of the Final Year Project Report is a black hardcover with the gold printed text of the following details in the given order: Name of the Institute; department; Type of document (i.e. Final Year Project Report); Title of the project; Name of the candidate; A statement of where the Final Year Project Report is submitted and the purpose; and Month and Year of completion/submission of the Final Year Project Report.

Title Page

The title page is the page that immediately follows the front cover; the title page contains the following details: The Institute logo with the name and motto of the institute embedded in the logo; Type of document (i.e. Final Year Project Report); Title of the project; Name of the candidate; Registration number of Candidate written in Brackets, Names, and titles of Supervisors, and Month and Year of completion/submission of the Final Year Project Report.

The title should be informative or descriptive indicating the focus and scope of the study while also being concise and catchy. It should contain the keywords of the report which should be less than 20 (twenty) words.

Declaration

This is where the candidate(s) declares that the work presented in the project report is his/ her/their work and has never been presented in any other University or institution for examination for an academic award e.g.

“We the undersigned students declare that this project report is unique and our original effort except where we have duly acknowledged”.

Approval

The approval is written on a separate page; the approval page gives a provision for the supervisors of the research to approve that the report was written under their supervision and is solely authored by the candidate(s). The supervisors shall approve a report that contains authentic work that is originally done by the author(s), has academic value and the project report is written following the Institutes’ approved formatting guidelines e.g.

“This is to certify that this project report has been prepared under my supervision. The report has been prepared in accordance with the guidelines governing the award of Diploma in at Uganda Institute of Information and Communications Technology (UICT).”

Dr. Keitesi Babara
(UICT)

.....
Supervisor

.....
Date

Dedication

The student may dedicate the work to some important person(s) in their life who have been instrumental in their educational journey. The student may mention the name(s) of persons to which the research is dedicated.

Acknowledgements

The acknowledgement entails recognition of mentors, colleagues, individuals, family members, sponsors, friends, and institutions that supported the research.

Table of Contents

A table of contents is generated to show the content of the report and the page numbers for easy reading and referencing.

It is a list of headings, subheadings with corresponding page numbers starting with the declaration page at (i), the title page is not numbered and does not appear in the table of contents.

*Every Appendix requires a title and is listed at the end of the table of contents in the order they appear in the report; **page numbers are not assigned to appendices** in the “Table of Contents”.*

List of Figures

This provides the captions of figures that appear in the project report with the corresponding pages where particular figures are found in the main body of the report. *Each figure must be captioned below it on, with figure numbers shown against each caption; the figures should be preferably numbered sequentially and according to chapters where they appear.*

List of Tables

This provides the list of captions of tables that appear in the research report with the corresponding pages where particular tables are found in the main body of the report for easy reading and referencing. *Each table must be captioned at the top of the table and with table numbers shown against it.* Tables should be preferably numbered sequentially and according to the chapter numbers where they appear.

Abbreviations /Acronyms

This page provided the abbreviation or acronyms commonly used in the Final Year project report. All acronyms or abbreviations that appear in the Final Year project report should be included in this section with their full meanings.

Abstract

The abstract is a summary of the project report which highlights the type of study, the problem addressed, main objective/purpose/aim and specific objectives, methods used in undertaking the research, key findings, conclusions, and recommendations.

The abstract has five parts:

- (a) Background/Introduction
- (b) Research Design or methodology
- (c) Findings/Results
- (d) Recommendations
- (e) Conclusion. (Stephan & Smith, 2019)

The abstract should preferably not exceed 300 words.

1.0 Chapter one:

1.1 Introduction

The purpose of the introduction section is to provide the rationale for the study. It establishes the research by detailing the importance of the focus area of the study, indicating the conditions of the existing situations relating to the area of study, highlighting the existing gap(s), and detailing the intervention of the study as a means of solving the problem or contributing to its solution. The introduction moves from the general discussion of the topic to the particular problem being investigated. Try to entice the reader to look forward to reading the rest of the chapter. You can do this using a quotation from an authority in the subject area.

1.2 Background to the study

The background of the study gives the reader an overview of the historical, conceptual, and context of the topic of the study. Explain the background of the problem about your topic. This includes the current state of affairs. Students need to articulate content that is relevant and related to the topic of study. Use citations from references consulted (APA format to be used when citing or referencing).

1.3 Statement of the Problem

The statement of the problem refers to the negative aspect that has been detected and needs a solution in the practical and theoretical world. What is the real and current problem? What effort has been put in place to solve the current problem? Highlight areas that continue to persist despite the current interventions. Then concisely state the problem one or two sentences. What is the likely consequence if this trend is not addressed?

1.4 General or Main Objective (Purpose of the Study)

The main objective refers to the general goal of the research by detailing what the research is supposed to accomplish. What do you expect to achieve in general terms. This also refers to the general intention of the project. It should spell out what the project is supposed to accomplish in a sentence or two. It must rhyme with the topic.

1.5 Specific Objectives

Specific objectives break down the general objective into smaller, logically connected parts that systematically address the various aspects of the problem. For each specific objective

you must have a method and tools that you will use to achieve it under methodology section (chapter 3). For any general objective related to development of a system/app, usually five specific objectives are expected i.e. a) To study b) To analyse C) To design, d) to develop, e) to test and validate For instance, consider the specific objectives below:

- a.** To study how KACITA staff interact with customers when booking for transport services.
- b.** To analyse the findings of the study in (a) above and determine user and system requirements for the web-based booking system
- c.** To design a web-based booking system (KACITA Online Booking System), based on the user and System requirements identified in (b) above.
- d.** To develop a web-based booking system (KACITA Online Booking System), based on the design in (c) above.
- e.** To test and validate the developed web-based booking system (KACITA Online Booking System) in (d) above.

1.6 Scope

The scope provides the depth and the boundary of investigation of the project in terms of sample size, technical scope, geographical scope, and time scope.

1.7 Significance of the study

1.7.1 To stakeholders:

This refers to the relevance or usefulness of the study to different stakeholders. What practical use might be made of the findings?

1.7.2 Market Survey and Viability

This part addresses the business dimension of the project, clearly outlining the place of the research in the National Development Goals.

The business model of the project should be well articulated so as to reveal the viability and sustenance of the project.

1.8 Justification/Rationale

The justification establishes why the research deserves to be undertaken by indicating its importance and potential threats the research if it is not done. It answers the question – Why? Would the world collapse if this work is not done?

2.0 Chapter two: Literature Review

The literature review constitutes an analysis of existing literature on the area of investigation. This section deals with a discussion of existing literature on the subject to reveal contributions made by earlier scholars, reveals weaknesses and gaps in existing knowledge, technologies, methods and practices, and lessons learned. The literature review should focus on the themes of the study and should inform the objectives and methodology of the study. It should be appropriately related to the topic and objectives hypotheses, methods, and research questions. This should be done objective-by-objective. The student will decide on sub-headings following research objectives. The citation should be following the APA citation format (for ICT students and IEEE citation format for engineering students), indicating author and year of publication following the approved format e.g. (Mugume, 2021). Quotations should include the page(s) where the quotations appear e.g. (Gonzaga, 2020, Pg. 3). The review should be critical and the use of current journal articles rather than textbooks is recommended.

2.1 Introduction

This gives an overview of the contents in this chapter, how the chapter is arranged.

2.2 Theoretical review

This will be the review of literature on the existing theories or concepts, which the candidate will have identified as being the guiding principle of the study. Show how the theories will be used in the study. These are existing theories or concepts supporting the system to be developed system.

This should reveal the global, regional and local context of the problem under study.

2.3 State-of-practice

Explain the current context or environment where the existing system is being used. It is not accurate to say there are no existing systems in place dealing with the identified problem. It's possible there is some sort of interventions but not sufficient to deal with the problem.

2.4 State-of-the-art /Similar existing systems

Explain the functionalities of the best benchmarked systems including the current system you would like to improve. It's better if these can be at least three systems studied i.e. Systems A, B, and C.

2.5 Comparative evaluation

Here students are required to compare the existing systems above (Systems A, B, and C.) and critique them using the table below to show the relevance and components of the proposed new system.

	System Functionalities				
Existing Systems studied	Does X	Does Y	Does Z	Does P	Does Q
System A	Y	N	Y	Y	N
System B	Y	Y	N	Y	Y
Current System(C)	N	Y	N	N	Y
Your new system	Y	Y	Y	Y	Y

2.5 Conclusion

This should be brief. At least a paragraph and should summarize what the student has discovered from the review of literature.

3.0 Chapter Three: Methodology

Under this chapter, show exactly how you have achieved each of the specific objectives stated in chapter one. Use present or past tense i.e. the researcher used (method/tool) to do because ... (gives why it's a preferable method compared to others). Also, explain briefly how you did (or how you have done) what you planned to do. For instance, to study existing systems and gather user and system requirements, some of the methods used include interviews, surveys, document reviews, and observation. The tools corresponding to these methods include an interview guide, questionnaire, document review checklist, observation checklist, software etc. In summary, the chapter includes steps, approaches, strategies, techniques, tools, the procedure used to achieve stated specific objectives.

Explain how each specific objectives was achieved

3.1 introduction

Give an introduction of the chapter

3.2 Study of how KACITA staff interact with customers when booking for transport services.

Explain the most preferable data collection method(s) used to achieve this specific objective.

- State the data collection tools of instruments that you used to achieve this specific objective.
- Briefly explain the procedure used to collect data for answering this specific objective.

3.3 Analysis of the findings of the study in (a) above and determine user and system requirements for the web-based booking system

- State the tools or instruments or software that you used to achieve this specific objective.
- What diagrams have been used to analyse data collected from objective a) above e.g. Use Case Diagrams, Flow Charts, and Data Flow Diagrams etc.

3.4 System Design a web-based booking system (KACITA Online Booking System), based on the user and System requirements identified in (b) above.

- State the tools or instruments or software that you used to achieve this specific objective.
- What UML diagrams have been used to design the desired system e.g. ERD, Data Dictionaries, etc.

3.5 System Development of a Web-Based Booking System (KACITA Online Booking System), based on the design in (c) above.

- State the tools or instruments or software that you used to achieve this specific objective.
- Specify the different parts (interfaces) of the designed system that will be developed and for what purpose.

3.5 Tests and validation of the developed web-based booking system (KACITA Online Booking System) in (d) above.

- Briefly explain the method and procedure used to test and validate the developed system
- Clearly indicate the input test data to be used and expected results (test data matrix)

4.0 System Analysis, Design and Development

4.1 System Analysis

Using relevant Unified Modeling Language (UML) tools, use case diagrams, flow charts, Data Flow Diagrams (DFDs) can be drawn to represent analysis information under section 4.1 above.

- ❑ This section is a continuation the corresponding methodology section
- ❑ Draw the corresponding diagrams used to analyse data e.g. Use Case Diagrams, Flow Charts, etc.
- ❑ For each diagram above, briefly explain its purpose in relation to other diagrams under this objective
- ❑ Clearly identify functional and non-functional requirements for the system to be developed.

4.2 System Design

Here, the definition of the components, modules, interfaces, and data that the system should satisfy are specified. Design methods may include: 1) Architectural design: to describe the views, models, behavior, and structure of the system. 2) Logical design: to represent the data flow, inputs, and outputs of the system. Examples: Entity Relationship Diagrams (ERDs), Data Dictionary, Table Structures, etc. Specifically:

- ❑ This section is a continuation the corresponding methodology section.
- ❑ Draw the corresponding diagrams used to design the desired system e.g. Data Flow Diagrams, ERD, Data Dictionary, etc.
- ❑ For each diagram above, briefly explain its purpose in relation to other diagrams under this objective

4.3 System Development

This section should explain the tools used to develop the system.

The system may have been developed by any applications software e.g. Android Studio, Java Programming Language, SQL, MySQL, PHP, etc. The student is at liberty to use any software of their choice as long they are able to demonstrate it.

- ❑ This sub-section is a continuation the corresponding methodology section
- ❑ Take screenshots of the different parts (interfaces) of the developed system showcasing its functionality.
- ❑ For each screenshot, briefly explain its purpose in relation to other diagrams under this objective.

4.4 System Interfaces

Under this section, screenshots showcasing the different functionalities of the developed system are briefly explained.

5.0 System Testing and Validation

5.1 System testing and validation

The student is expected to practically show how they carried out system validation including testing like functional testing, system testing, integration, and User acceptance testing. Using different inputs (matrix of test data) into the system and verifying the output, system validity should be tested. In each case, capture screenshots and give brief explanations of what was done, why it was done, and the output received. Use different stakeholders to test the system.

Specifically:

- ❑ This sub-section is a continuation the corresponding methodology section
- ❑ Take screenshots to showcase the output screens when input data is processed by the developed system.
- ❑ For each screenshot, briefly explain the result in relation to other diagrams under this objective.

6.0 Limitation, Challenges and Recommendations

6.1 Limitations

The student should show the weaknesses of the system developed explaining why such shortcomings were not addressed. For example students may show that a Library system is unable to support electronic payment of fines.

6.2 Challenges

The challenges show reasons for not achieving stated goals due to identifiable constraints. The examples would include failure to receive adequate data from the case study because of sensitivity and data protection mechanisms.

6.3 Recommendation

Recommendations are consistent with the purpose of the study, its objectives, the evidence presented, and the interpretations made. Such recommendations should be practical and achievable. Each of the above should be done objective by objective to give a logical flow.

References

This is a list of all works cited in the proposal and should be written according to the approved format. The approved format is APA Sixth or Seventh Edition

Appendices

The appendices are additional supplementary information that supports the content presented in the main body of the proposal. Each project report should have the following attachments as appendices: proposal (Budget, workplan), Questionnaire, Interview guide, document review checklist, observation checklist, and any other tools used.

Note: Budget & workplan should not be in the final report.